

Project Title: OmniMess – The Smart Hostel Network

1. The Core Problem

Our college has five hostels, each with its own mess facility. The current system faces three major challenges:

a. The Double Burden

Students claim a rebate (refund) for not eating at their home hostel but then secretly eat at another hostel's mess. As a result, the college incurs losses twice.

b. Massive Food Waste

Mess managers do not know the exact daily headcount. They prepare food for all students, but nearly 30% of the food often goes to waste.

c. Lack of Competitive Incentives

Since a student's administrative fee remains with their home hostel regardless of where they eat, messes have little incentive to improve food quality and services to attract or retain students.

2. The Smart Mess Solution

We are building a system where **"the money follows the student."**

The value of a daily meal is **₹150**, divided as follows:

- **₹96** – Variable Cost (raw food, vegetables, and ingredients)
- **₹54** – Administrative Charges (cook salaries, electricity, cleaning, and maintenance)

Rule 1: Pre-Booking Requirement

To receive a refund or switch hostels, students must notify the system through the mobile application before **10:00 PM on the previous day**.

If no action is taken, the system assumes that the student will eat at their home hostel. The student is charged **₹150**, and no rebate is provided.

Rule 2: Daily Rebate (Fairness Policy)

If a student informs the system in advance that they will be going home or eating elsewhere:

- The student continues paying the **₹54 administrative charge** to support hostel operations.
- The student receives a **₹96 refund** because the mess does not purchase food on their behalf.

Rule 3: Hostel Swap (Premium Choice)

If a student wishes to eat at a different hostel (e.g., Hostel B instead of their Home Hostel A):

Cost

The student pays an additional **₹50 convenience fee**, making the total cost **₹200**.

Money Transfer

The home hostel transfers the student's entire **₹150 meal allocation** to the guest hostel.

Reward

The guest hostel receives **₹200 in total** (₹150 transferred from the student's budget + ₹50 convenience fee). This additional revenue rewards hostels that provide better food and services.

3. Managing Crowd Capacity (30% Cap)

To prevent overcrowding at popular hostels:

- Each mess can accept guest bookings up to **30% of its resident capacity**.
 - For example, if a hostel has **200 residents**, the system allows a maximum of **60 guest bookings** for that day.
 - Once the limit is reached, the hostel is automatically locked for additional bookings.
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4. How the System Operates

Student App

Students can view the **crowd meter** of other hostels. If they wish to switch hostels, they pay the ₹50 convenience fee and reserve a slot before **10:00 PM**.

Manager Dashboard

At **10:01 PM**, every mess manager receives a preparation report.

Example:

- Total Residents: 200
- Absent Students: 40
- Guests Arriving: 20
- Meals to Prepare: 180

This allows managers to accurately estimate food requirements and reduce waste.

Gatekeeper System (QR Code + Facial Recognition)

At the mess entrance, students undergo **dual verification** using:

1. QR Code Scanning
2. Facial Recognition

Fraud Prevention

- If a student claims a rebate but still attempts to eat at the mess, the system instantly cancels the rebate and flags the student for a penalty.
- If a guest student has not pre-booked a meal, access is denied.

- Students cannot use another student's QR code because facial recognition must match the registered profile.
 - All entry attempts are logged for security and audit purposes.
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5. Key Benefits

For Students

- Receive refunds even for a single day of absence.
- Enjoy the flexibility to dine at other hostels by paying a small premium.
- Access real-time crowd information before making a booking.

For Managers

- Significantly reduce food waste.
- Know exactly how many meals need to be prepared each day.
- Improve operational planning and resource management.

For the College

- Eliminates the double-burden problem.
 - Prevents rebate fraud and unauthorized mess usage.
 - Encourages healthy competition among hostels.
 - Rewards hostels that provide better food and services.
 - Improves financial transparency and accountability.
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6. Project Goal

To build a secure, transparent, and efficient hostel mess management system that accurately manages meal bookings, rebates, hostel swaps, financial transfers, and access control while preventing fraud and minimizing food waste.